

Lecture 16

Super-Twisting Algorithm as a Controller, Observer and Differentiator

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Sliding mode control and its application

Outline

- 1 Super-Twisting Algorithm
- 2 Super-Twisting as a Controller
- 3 Super-Twisting as an Observer
- 4 Super-Twisting based differentiator

Super-Twisting Algorithm (STA)

The structure of STA is following

$$\begin{aligned}\dot{x}_1 &= -k_1 |x_1|^{\frac{1}{2}} \text{sign}(x_1) + x_2 \\ \dot{x}_2 &= -k_2 \text{sign}(x_1), \quad k_1 > 0, k_2 > 0\end{aligned}\tag{1}$$

- Using only x_1 information both the state variables x_1, x_2 will converge to zero, in finite time.
- Since the right-hand side is discontinuous, its solution will be understood in the Filippov sense.

Super-twisting with uncertainty

$$\begin{aligned}\dot{x}_1 &= -k_1 |x_1|^{\frac{1}{2}} \text{sign}(x_1) + x_2 + \rho_1(x, t) \\ \dot{x}_2 &= -k_2 \text{sign}(x_1) + \rho_2(x, t), \quad k_1 > 0, k_2 > 0\end{aligned}\tag{2}$$

Under some gain conditions on k_1 and k_2 , STA is robust and finite time stable against a bounded perturbation $\rho_1 = 0$ and $|\rho_2| \leq L$.

Lyapunov function for STA

Conditions for weak (non-strict) Lyapunov function

$$V(x) > 0, \quad \dot{V}(x) \leq 0, \quad \forall x \neq 0$$

Weak Lyapunov function for STA

$$V(x) = k_2|x_1| + \frac{1}{2}x_2^2$$

$$\dot{V}(x) = -k_1k_2|x_1|^{\frac{1}{2}}$$

- $\dot{V}(x)$ is only negative semi-definite.
- Finite time convergence can only be asserted by using a generalization of LaSalle's invariance principle for discontinuous systems.
- Above Lyapunov function is not able to provide robustness results and also estimation of the convergence time.

Lyapunov function for STA

Strong (strict) Lyapunov function conditions

$$V(x) > 0, \dot{V}(x) < 0, \forall x \neq 0 \quad (3)$$

- It assures robust stability.
- If $\dot{V} \leq -\gamma V^p, 0 < p < 1, \gamma > 0$ is satisfied. Then, finite time stability of trajectories w.r.t origin is ensured.

Define quadratic Lyapunov function $V(x) = \zeta^T P \zeta$ using a special state definition ζ .

Modified state is defined as

$$\zeta = \begin{bmatrix} \sqrt{|x_1|} \text{sign}(x_1) \\ x_2 \end{bmatrix}$$

Then we can find derivative of ζ as

$$\dot{\zeta} = \frac{1}{|x_1|^{\frac{1}{2}}} A \zeta \quad \text{where} \quad A = \begin{bmatrix} -\frac{1}{2}k_1 & \frac{1}{2} \\ -k_2 & 0 \end{bmatrix}$$

Strict Lyapunov function for STA

Strict Lyapunov function for STA was proposed by Moreno

$$V(x) = \zeta^T P \zeta = p_{11}|x_1| + 2p_{12}x_2|x_1|^{\frac{1}{2}}\text{sign}(x_1) + p_{22}x_2^2$$

where $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix}$ is a constant, symmetric, and positive definite matrix.

Some more discussion on the above Lyapunov function

- $V(x)$ is continuously differentiable, except on the set $S = \{(x_1, x_2) \in \mathbb{R}^2 | x_1 = 0\}$.
- From the set of differential inclusion of STA, it is easy to see that the trajectories of the system just cross the surface and cannot stay on it, except when the origin $x = 0$ has been reached.

By choosing matrix P as follows

$$P = \frac{1}{2} \begin{bmatrix} 4k_2 + k_1^2 & -k_1 \\ -k_1 & 2 \end{bmatrix}$$

we obtain the Lyapunov function as

$$V(x) = 2k_2|x_1| + \frac{1}{2}x_2^2 + \frac{1}{2}(k_1|x_1|^{1/2}\text{sign}(x_1) - x_2)^2.$$

Stability of proof

Derivative of Lyapunov function along the trajectories of unperturbed STA

$$\dot{V} = \zeta^T P \dot{\zeta} + \dot{\zeta}^T P \zeta = -|x_1|^{-\frac{1}{2}} \zeta^T Q \zeta$$

where P and Q are related by the Algebraic Lyapunov Equation (ALE)

$$A^T P + P A = -Q \quad (4)$$

- Since A is Hurwitz if and only if $k_1 > 0$, $k_2 > 0$, for every $Q = Q^T$ there exists a (unique) solution $P = P^T > 0$ of the ALE (4).
- Above analysis ensures the asymptotic stability of trajectories w.r.t origin. For finite time stability, more analysis is required.

Finite time stability analysis

- Using Rayleigh inequality, one can write

$$\lambda_{\min} \{P\} \|\zeta\|_2^2 \leq V(x) \leq \lambda_{\max} \{P\} \|\zeta\|_2^2$$

$$\lambda_{\min} \{Q\} \|\zeta\|_2^2 \leq \zeta^T Q \zeta \leq \lambda_{\max} \{Q\} \|\zeta\|_2^2$$

$$\Rightarrow -\zeta^T Q \zeta \leq -\lambda_{\min} \{Q\} \|\zeta\|_2^2 \quad (5)$$

where $\|\zeta\|_2^2 = |x_1| + x_2^2$ is the Euclidean norm of ζ .

- One can also note that

$$|x_1|^{\frac{1}{2}} \leq \|\zeta\|_2 \leq \frac{V^{\frac{1}{2}}}{\lambda_{\min}^{\frac{1}{2}} \{P\}} \quad (6)$$

- Using (5) time derivative of Lyapunov function is further written as

$$\dot{V} = -|x_1|^{-\frac{1}{2}} \zeta^T Q \zeta$$

$$\leq -|x_1|^{-\frac{1}{2}} \lambda_{\min} \{Q\} \|\zeta\|_2^2 \quad (7)$$

Time calculation

- Substituting $\|\zeta\|$ from (6) into (7)

$$\dot{V} \leq -\gamma V^{\frac{1}{2}}$$

$$\text{where } \gamma = \frac{\lambda_{\min}^{\frac{1}{2}}\{P\}\lambda_{\min}\{Q\}}{\lambda_{\max}\{P\}}$$

- Solution of the following differential equation (with initial condition $\nu(0) = \nu_0 \geq 0$)

$$\dot{\nu} = -\gamma \nu^{\frac{1}{2}}$$

is given by

$$\nu(t) = \left(\nu_0^{\frac{1}{2}} - \frac{\gamma}{2}t \right)^2 \quad (8)$$

- It follows from the **Comparison Principle** that $V(t) \leq \nu(t)$ when $V(x_0) \leq \nu_0$.
- From (8) one obtains that $V(x)$ and therefore $x(t)$, converges to zero in finite time and reaches that value at most after $T = \frac{2V^{\frac{1}{2}}(x_0)}{\gamma}$.

Super-Twisting as a Controller

Consider the linear time-invariant system of the following form

$$\dot{x}(t) = Ax(t) + B(u + \Psi_0)$$

Regular Form Transformation:-

Input matrix B may be partitioned as $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$ where $B_1 \in \mathbb{R}^{(n-m) \times m}$, $B_2 \in \mathbb{R}^{m \times m}$ with $\det B_2 \neq 0$.

$$T = \begin{bmatrix} I_{n-m} & -B_1 B_2^{-1} \\ 0 & B_2^{-1} \end{bmatrix}, \quad T_x = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

Transformed system

$$\begin{aligned} \dot{z}_1(t) &= A_{11}z_1(t) + A_{12}z_2(t) \\ \dot{z}_2(t) &= A_{21}z_1(t) + A_{22}z_2(t) + u + \Psi_0 \end{aligned} \tag{9}$$

where $z_1 \in \mathbb{R}^{n-m}$, $z_2 \in \mathbb{R}^m$ and

$$TAT^{-1} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad TB = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

Super-Twisting as a Controller

- **Design a sliding surface**

$$s = C_1 z_1 + z_2$$

where $s \in \mathbb{R}^{m \times 1}$ and $C_1 \in \mathbb{R}^{m \times (n-m)} > 0$.

- Note: For simplicity, we consider $m = 1$.
- Now, taking the first time derivative of the sliding surface

$$\begin{aligned}\dot{s} &= C_1 \dot{z}_1 + \dot{z}_2 \\ &= C_1 (A_{11} z_1(t) + A_{12} z_2(t)) + A_{21} z_1(t) + A_{22} z_2(t) + u + \Psi_0 \\ &= (C_1 A_{11} + A_{21}) z_1(t) + (C_1 A_{12} + A_{22}) z_2(t) + u + \Psi_0\end{aligned}\tag{10}$$

- Now expressing the system (9) into z_1 and s coordinate

$$\begin{aligned}\dot{z}_1(t) &= A_{11} z_1(t) + A_{12} (s - C_1 z_1(t)) \\ \dot{s} &= (C_1 A_{11} + A_{21}) z_1(t) + (C_1 A_{12} + A_{22}) (s - C_1 z_1(t)) + u + \Psi_0 \\ &= (C_1 A_{11} + A_{21} - C_1 A_{12} C_1 - A_{22} C_1) z_1(t) + (C_1 A_{12} + A_{22}) s + u + \Psi_0\end{aligned}\tag{11}$$

Super-Twisting as a Controller

Theorem

The control input u which is defined as

$$u = \nu - (C_1 A_{11} + A_{21} - C_1 A_{12} C_1 - A_{22} C_1) z_1(t) - (C_1 A_{12} + A_{22}) s \quad (12)$$

where

$$\begin{aligned} \nu &= -k_1 |s|^{\frac{1}{2}} \text{sign}(s) + w \\ \dot{w} &= -k_2 \text{sign}(s) \end{aligned} \quad (13)$$

with $k_1 > 0$, $k_2 > \|\Psi_0\|$ leads to the establishment s equal to zero in finite time.

Substituting u from eqn.(12) to (11), one can write

$$\begin{aligned} \dot{z}_1(t) &= A_{11} z_1(t) + A_{12}(s - C_1 z_1(t)) \\ \dot{s} &= -k_1 |s|^{\frac{1}{2}} \text{sign}(s) + w + \Psi_0 \\ \dot{w} &= -k_2 \text{sign}(s) \end{aligned} \quad (14)$$

Super-Twisting as a Controller

- Let $L = w + \Psi_0$ and $\dot{L} = \dot{w} + \dot{\Psi}_0 \Rightarrow \dot{L} = -k_2 \text{sign}(s) + \dot{\Psi}_0$. Hence

$$\dot{z}_1(t) = A_{11}z_1(t) + A_{12}(s - C_1z_1(t))$$

$$\dot{s} = -k_1|s|^{\frac{1}{2}}\text{sign}(s) + L$$

$$\dot{L} = -k_2\text{sign}(s) + \dot{\Psi}_0$$

- The Last two subequations form a Super-Twisting algorithm. Hence, by choosing $k_1 > 0$, $k_2 > \|\dot{\Psi}_0\|$ leads to the establishment s and L equal to zero in finite time.
- Therefore, after time T , dynamics of s has been collapsed, and the reduced order dynamics of system (14) after substituting $s = 0$, is given as

$$\dot{z}_1(t) = (A_{11} - A_{12}C_1)z_1(t)$$

- The stability and performance of the above reduced order system depend on (A_{11}, A_{12}) . Therefore, the design of C_1 depends on the controllability of the pair (A_{11}, A_{12}) .

Super-Twisting as an observer for the second order system

Consider the second-order system of the following form

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = f(x_1, x_2) + u$$

$$y = x_1$$

where $|f(x_1, x_2)| \leq f_0$.

The proposed Super-Twisting observer has the form

$$\dot{\hat{x}}_1 = \hat{x}_2 + L_1$$

$$\dot{\hat{x}}_2 = f(x_1, \hat{x}_2) + u + L_2 \quad (15)$$

where $\hat{x}_1$ and $\hat{x}_2$ are the state estimations, and output injections L_1 , L_2 are the correction variables given as

$$L_1 = k_1 |x_1 - \hat{x}_1|^{\frac{1}{2}} \text{sign}(x_1 - \hat{x}_1)$$

$$L_2 = k_2 \text{sign}(x_1 - \hat{x}_1) \quad (16)$$

Note: It is taken for the definiteness that at the initial moment $\hat{x}_1 = x_1$ and $\hat{x}_2 = 0$.

Super-Twisting as an observer for the second order system

Taking $e_1 = x_1 - \hat{x}_1$ and $e_2 = x_2 - \hat{x}_2$ we obtain the error equation

$$\begin{aligned}\dot{e}_1 &= e_2 - k_1 |e_1|^{\frac{1}{2}} \text{sign}(e_1) \\ \dot{e}_2 &= \hat{f}(x_1, x_2, \hat{x}_2) - k_2 \text{sign}(e_1)\end{aligned}\quad (17)$$

where $\hat{f}(x_1, x_2, \hat{x}_2) = f(x_1, x_2) - f(x_1, \hat{x}_2)$.

Suppose that the system states can be assumed bounded, then the existence is ensured of a constant f^+ , such that $|\hat{f}(x_1, x_2, \hat{x}_2)| < f^+$.

Theorem

Let the parameters of the observer (15), (16) be selected according to the Super-Twisting algorithm. Then, the error of the observer $e_1 = e_2 = 0$ converge in finite time, i.e., $(\hat{x}_1, \hat{x}_2) \rightarrow (x_1, x_2)$.

STA based ST differentiator

Our objective is to design a differentiator, which is able to calculate the first derivative of the given signal $x = f(t)$

We can define new variables x_1 and x_2 as follows

$$x = f(t) = x_1 \quad \text{and} \quad \dot{x} = \dot{f}(t) = x_2$$

This gives systems dynamics as

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \ddot{f}(t)$$

The proposed Super-Twisting differentiator has the form

$$\dot{\hat{x}}_1 = \hat{x}_2 + k_1 |x_1 - \hat{x}_1|^{\frac{1}{2}} \text{sign}(x_1 - \hat{x}_1)$$

$$\dot{\hat{x}}_2 = k_2 \text{sign}(x_1 - \hat{x}_1)$$

where $\hat{x}_1$ and $\hat{x}_2$ are the state estimations.

STA based ST differentiator

- Define the error variables as

$$e_1 = x_1 - \hat{x}_1$$

$$e_2 = x_2 - \hat{x}_2$$

- After differentiating error variables, we get

$$\dot{e}_1 = e_2 - k_1 |e_1|^{1/2} \text{sign}(e_1)$$

$$\dot{e}_2 = \ddot{f}(t) - k_2 \text{sign}(e_1)$$

- Above structure is similar to Super-Twisting provided $\ddot{f}(t)$ is bounded.
- Therefore, error dynamics will converge to zero in finite time, and $x_1 = \hat{x}_1 = f(t)$, $x_2 = \hat{x}_2 = \dot{f}$.

Thank You!